Patient Engagement Engine — Findings & Recommendation

Can we read a patient's communication style from their words to help clinicians tailor how they communicate?

June 9, 2026 · Prepared by Oscar · Validated on real, de-identified production data

> **Bottom line.** Detecting patient *personality* from text is not viable — we proved it on real visits and shelved it. We pivoted to reading the patient's *preferred decision-making style* (three groups: *let the doctor decide* / *decide together* / *decide themselves*), and that version **works**. It is promising but **not yet conclusively validated**: confirming it requires asking patients directly, which we don't do today. That gap points to a small, high-leverage opportunity.

What we found

Question	Verdict	Evidence (real production visits)
Detect **personality** (original idea)?	**Not viable**	Unsure on ~100% of patients and lumped almost everyone into 1–2 of 5 types. Personality isn't reliably readable from a conversation. Ruled out by evidence.
Detect **communication / decision style** (the pivot)?	**Works**	Now produces a usable, confident read for **~71%** of patients (was 0%), and classifies them realistically across all three groups — matching published medical norms.
Is it **accurate**?	**Promising**	Against an independent second read it agreed **68%** of the time — above the ~61% medical-literature benchmark and roughly 2× chance. Strong on the two common groups; **blind to the "let the doctor decide" group**. Not yet conclusive (see below).

What is still missing / not yet possible

- **No "answer key" yet.** True accuracy needs the patient's own answer to compare against. The system filled out the test; we never collected the answer key — because **we never ask patients how they prefer to decide.** The 68% above comes from a second AI reading the same transcripts, which is encouraging but not proof (two machines can share the same blind spots).
- **The natural place to ask — the intake questionnaire — is dormant.** The intake chatbot is fully built, but production shows it was only ever a **2025 pilot**: 34 intakes total (~1% of appointments), across 7 of 54 doctors, mostly loaded on a single day, and **none since July 2025**. It depended on an external system that stopped sending data. Nothing is broken — the data source went quiet.
- **Two known weak spots** to address: the blind spot on the "let the doctor decide" group, and an over-sensitive "patient is anxious" flag (fired for 71% of patients — not credible).

A high-leverage opportunity: capture patient name + email

To get the answer key we need to reach patients with one short question. Today we store only a patient's **name and date of birth** — no email — so we can't reliably reach them, and patient identity is fragile (duplicates and "Unknown" records from Zoom).

> Capturing a patient's **email** unlocks three things at once — and it **fits almost for free** into the calendar integration already on next sprint's roadmap (Acuity / Google Calendar), because those events already carry the patient's name and email:
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> - **Intake:** automatically send patients the link (today's missing "last mile").
> - **Zoom:** reliably match the right patient instead of name-guessing (fewer duplicates / "Unknown").
> - **Identity:** a dependable patient key, far better than name + date of birth.

Recommended next steps

When	Step	Effort
Now	Fix the two weak spots (anxiety flag + blind spot on the "let-the-doctor-decide" group)	Small (technical, cheap)
Next sprint	While building the planned calendar integration, also capture the patient's **email** (add it to the patient record)	Small *incremental* on planned work
In parallel	Decide where to ask the one decision-preference question (post-visit, or via a revived intake) and start collecting the "answer key"	Small–Medium
After data accrues	Run the real accuracy validation → final GO / NO-GO on the feature	Small (tooling already built)

Decisions we need from you

A. Keep investing in the communication-style engine? The pivot works and is promising; confirming it is low-to-medium cost. Stopping here loses the progress.

B. How do we collect the patient "answer key"? Quick path: a short post-visit question. Deeper path: make the intake ours (~1 week for a basic version where we create it and the doctor shares the link).

C. Capture patient name + email now? The strategic move — unlocks intake delivery, Zoom matching, and identity. Nearly free if it rides on the planned calendar work. Requires a quick compliance sign-off (email is sensitive patient data).

> **One honest caveat.** Building it is not the hard part — **adoption is.** Even during the pilot, only about 1 in 4 patients completed the intake. The bottleneck will be getting patients to engage, not the technology. Worth weighing before investing heavily.

Methodology: read-only analysis over de-identified production visit transcripts, scored with our existing Azure OpenAI (GPT-4.1) — no new vendor cost. Two random samples of 40 visits. No patient-identifying data was exposed. Detailed engineering findings are in the team's documentation.